

Carmen Lucca DiMario

(484) 467-2466

github.com/luccadimario

luccadimario@gmail.com

EDUCATION

Embry-Riddle Aeronautical University (ERAU)

Graduation:

Master of Science in Computer Science

GPA: 4.0

May 2027

Bachelor of Science in Computer Science

GPA: 3.97

May 2026

TECHNICAL SKILLS

Programming Languages: C, C++, Swift, Python, Bash, GNU ASM, Go, TypeScript, Java, SQL

Embedded Systems: STM32 Microcontrollers, Clang, ARMcc, NVIDIA Jetson Hardware, ROS2, TensorRT

Tools and Technologies: UNIX, Linux, macOS, Git, AWS, I2C/SPI (STM32)

PROFESSIONAL EXPERIENCE

Garmin: Embedded Software Development Intern

May 2026 - August 2026

- Migrated a legacy embedded library to a modernized codebase, identifying and resolving compatibility issues across compiler and runtime environments
- Implemented Clang compiler support as a replacement for legacy ARMcc toolchains, modernizing the build for avionics firmware
- Extended embedded library support for 64-bit architectures, resolving data type alignment and pointer-width issues to broaden platform compatibility
- Developed a system test suite for a proprietary avionics communication interface, validating real-time data transmission and protocol correctness

JRC Integrated Systems: RISE Intern/CO-OP Participant

June 2025 - May 2026

- Developed and optimized software tools for radiation-hardened semiconductor testing
- Upgraded Accuro's PETSc integration across a major version boundary to enable GPU-accelerated solvers for the differential equations at the core of its simulation
- Analyzed an advanced C/C++ codebase to characterize differences in Linear Energy Transfer (LET) through various substrates
- Rebuilt the Accuro test suite from hard-coded shell scripts into a Python framework with parallel execution, selectable test subsets, and configurable output, cutting a multi-hour run to roughly 25 minutes

PROJECT EXPERIENCE

Engineering Physics Propulsion Laboratory: AIRHOUND [Middleware Lead]

September 2024 - May 2026

- Designed a modular four-node ROS2 perception-to-control pipeline (detection, tracking, PINN prediction, PX4 bridge) communicating with a PX4 FMUv6X flight controller over Micro XRCE-DDS
- Diagnosed DDS image-transport serialization as the pipeline bottleneck and replaced it with direct V4L2 capture, cutting mean detection latency from 234 ms to 58.8 ms and eliminating all dropouts at 1280x720
- Deployed RF-DETR with TensorRT FP16 on a Jetson Orin at 15 FPS end-to-end, sustaining the 10 Hz PX4 offboard control loop

RoboRacer Autonomous Racing: Software Lead

August 2025 - May 2026

- Assembled and tuned an autonomous vehicle platform from components, integrating compute, sensors, and motor controller into a competition-ready system
- Brought up and configured the full ROS2 software stack on the onboard computer
- Deployed and validated an end-to-end autonomy stack from sensor input through planning to actuation under real competition conditions

PUBLICATIONS

DiMario, C., Bacha, R., & Butka, B. "Combating Senior Scams Using a Large Language Model-Created Rubric." ASSE '24, ACM, 2024. <https://doi.org/10.1145/3702138.3702140>

Malarchick, R., DiMario, C., et al. "Predictive Target Pursuit for Autonomous UAVs Using RF-DETR with Depth-Aware State Estimation and Physics-Informed Trajectory Prediction." Proc. SPIE 14030, 140300O (2026). <https://doi.org/10.1117/12.3094732>

EXTRACURRICULAR ACTIVITIES

- **ERAU Division II Cross-Country/Track:** Approximately 30 hours of training weekly
- **ERAU Cyber Rodeo 2026:** First Place Individual

<https://dimario.dev>